

GRAND FORKS AFB, ND, USA

WMO: 727575

Lat: 47.967N

Lon: 97.400W

Elev: 278

StdP: 98.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.0	(c) -26.3	(d) -32.1	(e) 0.2	(f) -28.4	(g) -29.4	(h) 0.3	(i) -25.9	(j) 14.4	(k) -12.5	(l) 12.8	(m) -11.8	(n) 3.5	(o) 310	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	31.7	22.3	30.0	21.3	28.4	20.3	24.7	28.7	23.1	27.7	22.0	26.4	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.7	26.7	22.1	17.3	25.8	20.4	15.6	24.2	76.3	29.1	70.4	27.1	65.6	25.8	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.5	(b) 11.1	(c) 9.7	(e) -32.4	(f) 34.9	(g) 3.8	(h) 1.3	(i) -35.1	(j) 35.9	(k) -37.3	(l) 36.6	(m) -39.4	(n) 37.4	(o) -42.1	(p) 38.3		
(a) 12.5	(b) 11.1	(c) 9.7	(e) -32.1	(f) 26.8	(g) 3.6	(h) 1.9	(i) -34.7	(j) 28.1	(k) -36.8	(l) 29.2	(m) -38.8	(n) 30.3	(o) -41.4	(p) 31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-13.8	-11.3	-3.7	4.7	12.5	18.3	20.9	19.8	14.8	6.8	-2.3	-9.8
	DBStd	13.49	8.07	7.76	7.80	5.77	4.84	3.79	3.00	3.28	4.70	5.12	6.43	7.23
	HDD10.0	3082	738	596	426	173	30	1	0	11	126	369	613	
	HDD18.3	5156	997	829	683	408	189	46	11	22	124	357	618	871
	CDD10.0	1196	0	0	2	14	107	248	337	305	155	28	1	0
	CDD18.3	230	0	0	0	0	8	44	89	68	19	1	0	0
	CDH23.3	2557	0	0	0	9	143	489	900	748	245	21	0	0
	CDH26.7	775	0	0	0	2	36	143	271	246	73	4	0	0
(14) Wind	WSAvg	4.7	5.1	5.2	5.3	5.1	5.0	4.3	3.8	3.8	4.4	4.9	4.9	4.9
(15) Precipitation	PrecAvg	541	11	11	19	27	81	96	93	73	50	45	22	14
	PrecMax	654	29	35	58	67	228	200	191	143	136	152	101	33
	PrecMin	360	0	0	0	6	23	25	25	18	7	1	0	1
	PrecStd	79	9	10	17	15	46	40	45	27	31	35	23	9
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	7.7	17.7	24.1	30.5	33.7	32.9	34.0	32.0	26.1	16.1	7.0
		MCWB	2.9	4.4	11.7	13.1	18.3	22.1	23.9	24.0	21.7	16.3	11.0	3.4
	2%	DB	3.0	4.1	12.9	21.0	27.2	30.0	31.1	31.4	28.9	21.3	12.2	3.9
		MCWB	1.2	2.6	8.9	11.9	16.4	20.4	23.3	21.9	19.3	13.7	7.4	1.8
	5%	DB	0.3	2.3	9.1	17.6	24.4	28.0	29.6	29.3	26.3	17.8	9.1	2.0
		MCWB	-0.5	1.3	6.2	10.0	14.8	19.2	22.4	21.1	17.7	12.0	5.9	0.6
	10%	DB	-2.3	0.1	5.9	14.8	22.1	26.2	27.9	27.5	23.7	15.1	6.3	-0.1
		MCWB	-3.0	-0.6	4.1	8.7	13.8	18.5	21.1	19.9	16.8	10.7	3.9	-1.0
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	5.3	13.5	14.2	20.2	25.2	27.2	26.7	23.2	17.7	11.1	4.0
		MCDB	5.2	6.8	15.9	20.6	25.5	29.1	29.1	29.9	28.6	22.8	15.3	6.1
	2%	WB	1.4	2.9	9.3	12.3	18.6	22.9	25.2	23.9	21.0	15.2	8.2	2.2
		MCDB	2.9	4.1	12.7	19.0	24.3	27.5	28.9	28.2	25.9	19.7	11.3	3.7
	5%	WB	-0.4	1.3	6.2	10.7	16.8	21.0	23.7	22.5	19.3	12.9	6.1	0.6
		MCDB	0.6	2.1	8.8	16.6	21.2	25.3	27.7	27.2	23.9	17.2	8.9	1.7
	10%	WB	-2.9	-0.6	3.9	9.1	15.2	19.8	22.5	21.2	17.9	10.8	3.9	-1.1
		MCDB	-2.3	0.2	5.9	14.3	20.0	24.1	26.6	25.5	22.2	14.7	6.1	-0.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.7	9.7	11.9	13.1	12.0	12.2	13.2	12.8	10.8	9.3	8.5
		MCDBR	10.5	10.2	14.0	17.6	17.0	14.8	13.8	15.4	16.5	16.0	13.8	10.6
	5% WB	MCWBR	9.3	8.4	9.7	9.7	8.6	7.3	7.5	7.3	8.2	9.2	9.4	8.8
		MCWBR	10.6	9.5	13.4	16.5	13.3	12.5	12.4	13.4	13.8	14.2	13.2	10.1
(40) Clear-Sky Solar Irradiance	taub	0.283	0.300	0.300	0.353	0.368	0.381	0.381	0.383	0.348	0.305	0.275	0.261	
		taud	2.287	2.268	2.445	2.353	2.373	2.366	2.371	2.366	2.455	2.546	2.517	2.424
	Ebn at Noon	786	856	926	898	894	881	875	858	860	854	814	786	
		Edh at Noon	74	95	94	114	116	118	116	112	93	72	59	57
(44) All-Sky Solar Radiation	RadAvg	1.58	2.78	3.91	4.87	5.49	5.99	6.34	5.29	3.88	2.40	1.42	1.09	
	RadStd	0.17	0.25	0.41	0.44	0.35	0.45	0.27	0.36	0.27	0.35	0.17	0.12	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page